

Amanda Mai

amandamimai@gmail.com | 1 (714)-804-9871 | <https://www.linkedin.com/in/amanda-mai-b2129a256/> | <https://github.com/moomoonda>

PROFESSIONAL SUMMARY

Computer Engineering graduate from UCLA pursuing Hardware and Embedded Engineering roles in consumer electronics. Experienced in embedded firmware development, digital hardware design, and hardware-software co-design across ESP32, FPGA, and VLSI platforms. Skilled in low-level C/C++, FreeRTOS, Verilog, and LTSpice, with coursework spanning signal processing, electronic circuits, and feedback control systems.

EDUCATION

California State Polytechnic University, Pomona

M.S. Electrical Engineering.

Pomona, CA

Expected Fall 2026

University of California, Los Angeles (UCLA)

B.S. Computer Engineering. GPA: 3.43/4.00.

Los Angeles, CA

June 2026

Relevant Coursework: Data Communications and Telecommunications, Engineering Electromagnetics, Digital Signal Processing, Speech and Image Processing, Systems and Signals, Electronic Circuits, Probability and Statistics, Feedback Control Systems

EXPERIENCE

Software Engineering Intern

Jun. 2025 – Sept. 2025

TrueCx

- Developed Python diagnostic tools for system monitoring, focusing on latency and hardware-resource utilization.
- Built QA test matrices and Jira bug-tracking workflows to improve system reliability.
- Optimized AI scoring pipeline consistency using structured prompts and Postgres-backed validation.

PROJECTS

Embedded Missing Item Tag Tracker (ESP32)

Mar. 2026 – June 2026

- Designed dual-core ESP32 firmware using FreeRTOS to separate interrupt handling from file-system operations.
- Implemented persistent storage using LittleFS and cJSON for power-loss recovery.
- Reduced detection latency through O(1) lookup-table-based event processing.

4-Bit Absolute Value Detector (VLSI / Circuit Design)

Jan. 2025 – Mar. 2025

- Designed a 4-bit arithmetic circuit in LTSpice, comparing Static CMOS vs. Transmission Gate logic architectures.
- Achieved 70% energy reduction via gate sizing and Voltage Scaling, accepting a 40% delay trade-off.
- Conducted PPA (Power, Performance, Area) analysis and gate sizing to optimize noise margins, signal integrity, and low-power suitability for embedded applications.

FPGA Obstacle Game (Verilog, Basys 3)

Nov. 2024 – Dec. 2024

- Developed a real-time hardware controller using FSMs for game logic and physics on an Artix-7 FPGA.
- Interfaced 7-segment displays and GPIO-based button inputs with hardware-level debouncing logic.
- Synchronized clock domains to ensure stable game-state transitions and responsive user interaction.

Out-of-Order Processor Simulator (Computer Architecture)

Sept. 2025 – Dec. 2025

- Developed a cycle-accurate simulator modeling fetch, dispatch, and retirement stages with configurable functional units.
- Implemented Tomasulo's Algorithm, including Reservation Stations and Register Renaming, to handle RAW/WAR/WAW data hazards.
- Analyzed IPC throughput by simulating hardware bottlenecks and structural hazards within the execution pipeline.

Speech Signal Processing (Digital Signal Processing)

Sept. 2025 – Dec. 2025

- Built a speech-classification pipeline using MFCC, DFT, LPC, ZCR, spectral centroid, and RMS features.
- Evaluated filtering methods, including low-pass, high-pass, median, Savitzky-Golay, and pre-emphasis, across clean and noisy datasets.
- Achieved 81% accuracy on clean speech and identified Savitzky-Golay filtering as the most robust under noise.

SKILLS

Hardware and Embedded: Verilog, FPGA (Basys 3), ESP32, FreeRTOS, Digital Logic Design, LTSpice Simulation, Oscilloscopes, Multimeters

Programming: C++, C, Python, MATLAB, Assembly

Architecture: RISC-V, x86, ARM, CMOS Logic, Computer Arithmetic

Software and Tools: Git, Linux, Jira